A HYBRID ARCHITECTURE FOR 2D GAME DEVELOPMENT: THE CASE OF ASTEROID MINER

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Introduction

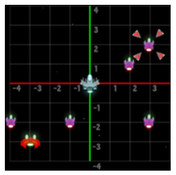
Modern video games often require efficient decision-making in resource-constrained environments, especially in scenarios involving resource management and navigation. This project explores how classical algorithms can be applied to optimize these processes in a 2D game environment, providing both real-time performance and analytical insights.

Objective

To design a 2D game that maximizes resource collection under fuel constraints by applying and comparing different algorithmic strategies, including greedy and backtracking approaches.

Methodology

Background



The development of efficient decision-making systems is a fundamental challenge in modern video game design, particularly in resource management. Asteroid Miner is a 2D environment where a mining unit must navigate a field of coordinates to collect resources under strict fuel constraints.

Algorithmic Logic

Greedy Algorithm

Used for real-time asteroid selection. It prioritizes targets based on the Value-to-Distance ratio, offering high speed and low computational cost for immediate decisions.

Backtracking Model

Selected for optimal route planning. It systematically explores possible sequences of asteroid visits to ensure the path maximizes total resources without exceeding the fuel limit.

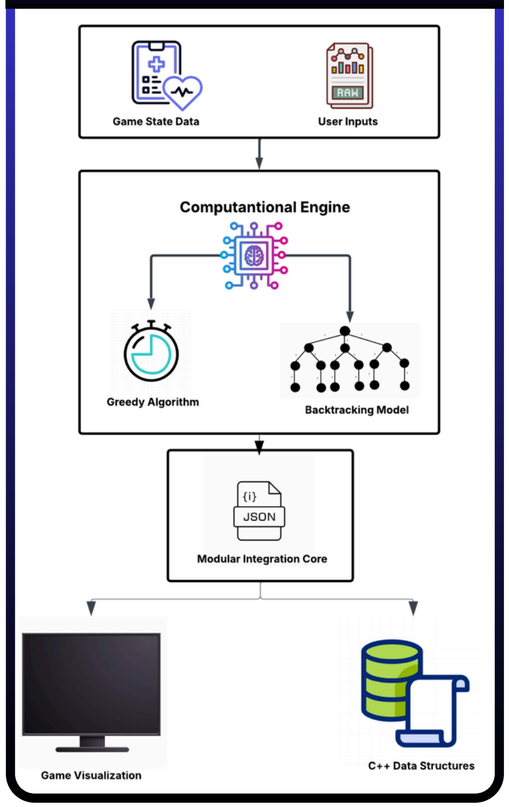
Results

The preliminary results of the system demonstrate clear differences between the implemented algorithmic approaches in terms of efficiency and optimization. The greedy algorithm, based on the value-distance relationship, provides fast and computationally efficient solutions; however, tbut does not always achieve the maximum possible resource collection.

On the other hand, the backtracking algorithm guarantees the optimal solution by exhaustively exploring all feasible routes under the fuel constraint. This ensures maximum resource collection and a safe return to base. However, this approach has a significantly higher computational cost.

Although they function differently and yield distinct results, we believe that both algorithms will be very useful for the development and efficiency of our game.

Architect



Conclusion

The Asteroid Miner project demonstrates that a hybrid architecture successfully balances real-time performance with global optimization by integrating Python's rapid UI capabilities with C++'s efficient data management.